

The Silent Tax: Typo-Robustness Analysis
(GSM8K_FIX-SPELLCHECK_GRID)
DeepSeek-R1-Distill-Qwen-7B

All tables are *answered-only*: traces that hit the token cap before writing a final answer are excluded from the metrics. Configs are named `typo{rate}_real{ratio}`: *rate* = % of words corrupted, *real* = % of typos that form real words. Baseline is the clean input.

Accuracy and Flips

Table 1: Accuracy per config. n = questions; $n_answered$ = questions that produced a final answer (not truncated at the token cap). $acc_answered$ = correct / $n_answered$; acc_strict = correct / n (truncated counted as wrong); ci_lo/ci_hi = 95% bootstrap CI on acc_strict .

config	n	n_answered	acc_answered	acc_strict	ci_lo	ci_hi
clean	1319	1306	0.8407	0.8324	0.8112	0.8537
rate25_real70	500	487	0.7618	0.742	0.702	0.78
rate75_real20	500	491	0.5825	0.572	0.53	0.616
rate50_real20	500	487	0.6756	0.658	0.616	0.702
rate25_real50	500	485	0.7299	0.708	0.668	0.748
rate75_real40	500	493	0.5456	0.538	0.494	0.582
rate50_real60	500	489	0.6196	0.606	0.564	0.648
rate25_real10	500	485	0.7732	0.75	0.712	0.784
rate25_real0	500	483	0.7867	0.76	0.724	0.798
rate50_real0	500	486	0.714	0.694	0.654	0.732
rate25_real30	500	487	0.7495	0.73	0.692	0.768
rate75_real60	500	485	0.5155	0.5	0.458	0.546
rate50_real40	500	489	0.6646	0.65	0.608	0.694
rate75_real30	500	484	0.5496	0.532	0.49	0.578
rate50_real10	500	486	0.6934	0.674	0.634	0.714
rate25_real60	500	484	0.7479	0.724	0.682	0.762
rate25_real40	500	488	0.7377	0.72	0.68	0.756
rate75_real0	500	480	0.6562	0.63	0.588	0.674
rate75_real10	500	490	0.6286	0.616	0.576	0.658
rate50_real30	500	486	0.6687	0.65	0.608	0.69
rate50_real70	500	489	0.6053	0.592	0.548	0.638
rate75_real50	500	492	0.4939	0.486	0.444	0.528
rate50_real50	500	492	0.628	0.618	0.576	0.66
rate75_real70	500	490	0.4776	0.468	0.424	0.512
rate25_real20	500	481	0.7651	0.736	0.696	0.774
typo30_real0	500	486	0.7778	0.756	0.716	0.792
typo30_real10	500	487	0.7906	0.77	0.73	0.806
typo30_real20	500	489	0.7791	0.762	0.724	0.798
typo30_real30	500	476	0.7563	0.72	0.682	0.758
typo30_real40	500	487	0.77	0.75	0.712	0.788
typo30_real50	500	484	0.7438	0.72	0.68	0.756
typo30_real60	500	485	0.734	0.712	0.672	0.752
typo30_real70	500	483	0.7288	0.704	0.662	0.744

Table 2: Answered-only flips vs clean, paired by question over n_both (answered in both conditions). r2w / w2r = questions correct on clean but wrong on this config / wrong on clean but correct on this config; r2w_ratio = r2w / n_both, w2r_ratio = w2r / n_both; net_flip_ratio = r2w_ratio - w2r_ratio = (r2w - w2r) / n_both, the net rate at which this config worsens answers relative to clean.

config	n_both	r2w	w2r	r2w_ratio	w2r_ratio	net_flip_ratio	mcnemar_b	mcnemar_c	mcnemar_stat	mcnemar_p
rate25_real70	485	64	27	0.132	0.0557	0.0763	64	27	14.242	0.00016076
rate75_real20	489	139	17	0.2843	0.0348	0.2495	139	17	93.853	0.0
rate50_real20	484	105	26	0.2169	0.0537	0.1632	105	26	46.443	0.0
rate25_real50	482	68	19	0.1411	0.0394	0.1017	68	19	26.483	2.7e-07
rate75_real40	490	160	17	0.3265	0.0347	0.2918	160	17	113.921	0.0
rate50_real60	487	127	20	0.2608	0.0411	0.2197	127	20	76.435	0.0
rate25_real10	483	57	27	0.118	0.0559	0.0621	57	27	10.012	0.00155532
rate25_real0	482	46	16	0.0954	0.0332	0.0622	46	16	13.565	0.0002305
rate50_real0	484	84	24	0.1736	0.0496	0.124	84	24	32.231	1e-08
rate25_real30	486	66	22	0.1358	0.0453	0.0905	66	22	21.011	4.57e-06
rate75_real60	483	171	15	0.354	0.0311	0.323	171	15	129.167	0.0
rate50_real40	488	103	19	0.2111	0.0389	0.1721	103	19	56.467	0.0
rate75_real30	482	154	14	0.3195	0.029	0.2905	154	14	115.006	0.0
rate50_real10	484	92	23	0.1901	0.0475	0.1426	92	23	40.209	0.0
rate25_real60	482	67	24	0.139	0.0498	0.0892	67	24	19.385	1.069e-05
rate25_real40	486	72	19	0.1481	0.0391	0.1091	72	19	29.714	5e-08
rate75_real0	478	114	28	0.2385	0.0586	0.1799	114	28	50.88	0.0
rate75_real10	488	125	23	0.2561	0.0471	0.209	125	23	68.926	0.0
rate50_real30	484	100	16	0.2066	0.0331	0.1736	100	16	59.388	0.0
rate50_real70	487	129	17	0.2649	0.0349	0.23	129	17	84.39	0.0
rate75_real50	489	179	14	0.3661	0.0286	0.3374	179	14	139.358	0.0
rate50_real50	489	121	20	0.2474	0.0409	0.2065	121	20	70.922	0.0
rate75_real70	487	189	16	0.3881	0.0329	0.3552	189	16	144.312	0.0
rate25_real20	480	61	25	0.1271	0.0521	0.075	61	25	14.244	0.00016056
typo30_real0	485	57	25	0.1175	0.0515	0.066	57	25	11.72	0.00061848
typo30_real10	486	51	25	0.1049	0.0514	0.0535	51	25	8.224	0.00413472
typo30_real20	486	57	29	0.1173	0.0597	0.0576	57	29	8.477	0.00359715
typo30_real30	475	59	19	0.1242	0.04	0.0842	59	19	19.5	1.006e-05
typo30_real40	485	59	25	0.1216	0.0515	0.0701	59	25	12.964	0.00031749
typo30_real50	482	69	24	0.1432	0.0498	0.0934	69	24	20.817	5.05e-06
typo30_real60	484	72	22	0.1488	0.0455	0.1033	72	22	25.543	4.3e-07
typo30_real70	481	79	25	0.1642	0.052	0.1123	79	25	27.01	2e-07

McNemar test (used in the flips table above).

- **What it is:** a paired significance test for before/after binary outcomes. It looks only at the questions that *changed* (the discordant pairs: b = right→wrong, c = wrong→right) and ignores those that stayed the same.
- **Null hypothesis:** typos break correct answers as often as they fix wrong ones ($b = c$, i.e. no net effect).
- **Statistic:** $(|b - c| - 1)^2 / (b + c)$ (column mcnemar_stat); mcnemar_p is its chi-square p -value with 1 degree of freedom.
- **When we reject it:** when mcnemar_p < 0.05 — the right→wrong vs wrong→right imbalance is too large to be chance.

Reasoning Length

Table 3: Absolute generated tokens per config (answered-only).

config	n_answered	mean_tok	median_tok	p90_tok	max_tok
clean	1306	589.0	449.5	1110	3966
rate25_real70	487	646.4	469	1209	3972
rate75_real20	491	619.0	462	915	4084
rate50_real20	487	635.6	470	1038	4096
rate25_real50	485	644.6	468	1191	4096
rate75_real40	493	601.9	459	946	4096
rate50_real60	489	576.0	450	880	3508
rate25_real10	485	695.2	487	1395	4096
rate25_real0	483	637.3	458	1112	4096
rate50_real0	486	620.9	459.5	984	4096
rate25_real30	487	640.4	468	1260	4096
rate75_real60	485	585.2	465	845	4096
rate50_real40	489	610.5	456	1001	4096
rate75_real30	484	596.3	468.0	941	4096
rate50_real10	486	622.9	457.0	1179	4096
rate25_real60	484	660.3	466.0	1254	3764
rate25_real40	488	661.6	462.0	1429	4096
rate75_real0	480	606.0	460.0	863	4096
rate75_real10	490	666.6	467.0	1236	4096
rate50_real30	486	624.2	467.5	1131	4096
rate50_real70	489	584.6	454	975	4096
rate75_real50	492	588.3	461.0	850	4096
rate50_real50	492	583.8	461.0	837	4096
rate75_real70	490	570.3	458.5	768	4096
rate25_real20	481	666.5	482	1263	4096
typo30_real0	486	632.4	475.5	1077	4096
typo30_real10	487	657.4	463	1278	4096
typo30_real20	489	677.3	476	1340	4096
typo30_real30	476	616.4	459.0	1071	3743
typo30_real40	487	642.6	460	1197	4096
typo30_real50	484	636.4	456.5	1174	3980
typo30_real60	485	656.6	456	1326	4096
typo30_real70	483	606.0	456	1082	4096

Table 4: Mean generated tokens for correct vs wrong answers. $\text{wrong_over_correct} = \text{tok_wrong} / \text{tok_correct}$ (mean tokens of wrong answers divided by mean tokens of correct answers).

config	tok_correct	tok_wrong	wrong_over_correct
clean	582.6	622.8	1.0689
rate25_real70	629.9	699.1	1.1098
rate75_real20	595.8	651.3	1.0931
rate50_real20	621.0	666.1	1.0726
rate25_real50	607.0	746.0	1.2289
rate75_real40	581.9	626.0	1.0758
rate50_real60	569.7	586.2	1.0289
rate25_real10	660.6	813.4	1.2313
rate25_real0	620.5	699.3	1.127
rate50_real0	624.0	613.0	0.9823
rate25_real30	618.9	704.8	1.1387
rate75_real60	529.9	644.0	1.2153
rate50_real40	593.6	644.0	1.0849
rate75_real30	579.3	617.1	1.0653
rate50_real10	636.7	591.7	0.9294
rate25_real60	639.9	720.9	1.1266
rate25_real40	649.7	695.0	1.0698
rate75_real0	590.3	635.9	1.0773
rate75_real10	614.0	755.6	1.2306
rate50_real30	626.9	618.8	0.9871
rate50_real70	595.6	567.8	0.9533
rate75_real50	538.1	637.2	1.1843
rate50_real50	544.5	650.2	1.194
rate75_real70	529.7	607.3	1.1465
rate25_real20	649.7	721.0	1.1098
typo30_real0	611.4	705.8	1.1545
typo30_real10	642.8	712.5	1.1085
typo30_real20	627.7	852.2	1.3577
typo30_real30	619.5	606.9	0.9796
typo30_real40	612.3	744.2	1.2155
typo30_real50	614.7	699.6	1.138
typo30_real60	652.2	668.8	1.0253
typo30_real70	617.7	574.6	0.9304

Self-doubt

The self-doubt markers are curated words grouped into two *banks*, each matched as a case-insensitive whole-word pattern in the reasoning text. Density is matches per 1000 reasoning words.

- **second_guess** — the proposal’s canonical self-correction words plus direct reversals: *wait, hmm, actually, reconsider, but wait, hold on, on second thought, let me recheck, double-check*.
- **uncertainty** — hedging / “not certain” words: *maybe, perhaps, possibly, might be, could be, not sure, I guess, presumably, or maybe / alternatively, unclear / confused*.

Example reasoning: “Wait, maybe the total is 12... *hmm, actually let me recheck*” contributes *wait, hmm, actually* (second_guess) and *maybe* (uncertainty). A separate *typo-noticing* bank (*typo, misspelled, doesn’t make sense, ...*) is treated as repair behaviour, not self-doubt.

Table 5: Doubt-marker density per config: second_guess and uncertainty, per 1k words and per trace.

config	n_answered	second_guess_per_1k	second_guess_per_trace	uncertainty_per_1k	uncertainty_per_trace	total_per_1k	total_per_trace
clean	1306	7.15	1.494	3.47	0.725	10.62	2.219
rate25_real70	487	8.554	2.133	7.525	1.877	16.08	4.01
rate75_real20	491	7.57	1.678	9.315	2.065	16.885	3.743
rate50_real20	487	8.658	2.076	7.904	1.895	16.562	3.971
rate25_real50	485	8.52	2.115	7.059	1.753	15.579	3.868
rate75_real40	493	6.252	1.331	10.074	2.144	16.326	3.475
rate50_real60	489	7.351	1.436	7.634	1.491	14.984	2.926
rate25_real10	485	9.811	2.746	6.593	1.845	16.404	4.592
rate25_real0	483	8.854	2.174	5.65	1.387	14.504	3.561
rate50_real0	486	7.29	1.667	5.949	1.36	13.239	3.027
rate25_real30	487	8.535	2.105	5.937	1.464	14.472	3.569
rate75_real60	485	6.626	1.307	9.375	1.849	16.001	3.157
rate50_real40	489	7.765	1.697	8.121	1.775	15.886	3.472
rate75_real30	484	7.093	1.446	6.86	1.399	13.954	2.845
rate50_real10	486	8.421	1.977	9.297	2.183	17.717	4.16
rate25_real60	484	8.04	2.052	8.137	2.076	16.177	4.128
rate25_real40	488	8.918	2.328	8.706	2.273	17.623	4.6
rate75_real0	480	7.124	1.51	8.854	1.877	15.978	3.388
rate75_real10	490	8.944	2.324	11.834	3.076	20.778	5.4
rate50_real30	486	8.004	1.866	7.572	1.765	15.576	3.632
rate50_real70	489	7.044	1.466	7.506	1.562	14.55	3.029
rate75_real50	492	6.146	1.213	8.73	1.724	14.875	2.937
rate50_real50	492	5.907	1.187	6.454	1.297	12.361	2.484
rate75_real70	490	5.984	1.112	9.575	1.78	15.559	2.892
rate25_real20	481	9.679	2.561	7.895	2.089	17.574	4.651
typo30_real0	486	8.276	1.951	4.461	1.051	12.737	3.002
typo30_real10	487	9.146	2.363	7.398	1.912	16.545	4.275
typo30_real20	489	8.795	2.325	5.995	1.585	14.79	3.91
typo30_real30	476	8.279	1.889	7.192	1.641	15.471	3.529
typo30_real40	487	9.161	2.279	7.345	1.828	16.506	4.107
typo30_real50	484	8.74	2.124	7.814	1.899	16.554	4.023
typo30_real60	485	8.575	2.151	6.988	1.753	15.563	3.903
typo30_real70	483	8.189	1.826	5.432	1.211	13.621	3.037

Table 6: Per-marker density (markers per 1000 reasoning words): clean vs pooled-typo. discrimination = typo minus clean (larger = more typo-responsive). All bank markers, ranked by discrimination.

marker	category	clean_per_1k	typo_per_1k	discrimination
maybe	uncertainty	1.484	3.908	2.423
wait	second_guess	3.108	4.039	0.932
or maybe/alt	uncertainty	1.096	1.798	0.702
perhaps	uncertainty	0.341	0.842	0.501
unclear/confused	uncertainty	0.246	0.543	0.298
hmm	second_guess	1.656	1.827	0.171
might be	uncertainty	0.055	0.194	0.139
but wait	second_guess	0.322	0.415	0.093
not sure	uncertainty	0.022	0.101	0.079
actually	second_guess	0.191	0.239	0.048
I assume	uncertainty	0.022	0.064	0.042
not entirely	uncertainty	0.004	0.023	0.02
could be	uncertainty	0.07	0.081	0.011
presumably	uncertainty	0.004	0.006	0.002
reconsider	second_guess	0.0	0.001	0.001
on second thought	second_guess	0.0	0.0	0.0
possibly	uncertainty	0.004	0.002	-0.001
I guess	uncertainty	0.125	0.074	-0.051
wait/actually no	second_guess	0.264	0.209	-0.055
hold on	second_guess	0.48	0.411	-0.069
let me recheck	second_guess	0.55	0.473	-0.076
double-check	second_guess	0.579	0.489	-0.09

Table 7: Doubt density (markers per 1000 reasoning words) split by final outcome, per category (2g = second.guess, un = uncertainty, tot = both). For a group, density = total markers in that group / total reasoning words in that group x 1000. *_correct = over correct-answer traces; *_wrong = over wrong-answer traces; *_wrong_over_correct = _wrong / _correct.

config	2g_correct	2g_wrong	2g_wrong_over_correct	un_correct	un_wrong	un_wrong_over_correct	tot_correct	tot_wrong	tot_wrong_over_correct
clean	7.328	6.151	0.839	3.176	5.114	1.61	10.504	11.265	1.072
rate25_real70	9.031	7.109	0.787	5.692	13.088	2.299	14.723	20.197	1.372
rate75_real20	7.569	7.57	1.0	8.818	9.986	1.132	16.387	17.556	1.071
rate50_real20	8.649	8.676	1.003	7.477	8.809	1.178	16.126	17.485	1.084
rate25_real50	8.66	8.203	0.947	5.75	10.017	1.742	14.41	18.219	1.264
rate75_real40	6.718	5.689	0.847	9.05	11.315	1.25	15.767	17.003	1.078
rate50_real60	7.635	6.845	0.897	7.389	8.069	1.092	15.024	14.914	0.993
rate25_real10	9.765	9.939	1.018	5.43	9.8	1.805	15.195	19.739	1.299
rate25_real0	8.898	8.698	0.977	5.183	7.312	1.411	14.081	16.01	1.137
rate50_real0	7.983	5.11	0.64	6.453	4.364	0.676	14.437	9.475	0.656
rate25_real30	8.582	8.408	0.98	5.026	8.408	1.673	13.608	16.815	1.236
rate75_real60	5.907	7.204	1.219	7.009	11.277	1.609	12.917	18.481	1.431
rate50_real40	8.789	5.65	0.643	6.887	10.668	1.549	15.676	16.318	1.041
rate75_real30	7.424	6.669	0.898	4.883	9.401	1.925	12.307	16.07	1.306
rate50_real10	9.268	5.945	0.642	8.927	10.378	1.163	18.194	16.324	0.897
rate25_real60	8.199	7.592	0.926	7.014	11.295	1.61	15.213	18.887	1.241
rate25_real40	9.001	8.673	0.964	8.233	10.098	1.226	17.234	18.77	1.089
rate75_real0	7.89	5.569	0.706	8.301	9.977	1.202	16.191	15.546	0.96
rate75_real10	8.248	9.86	1.195	7.792	17.155	2.202	16.041	27.015	1.684
rate50_real30	8.703	6.27	0.72	7.044	8.883	1.261	15.747	15.153	0.962
rate50_real70	7.719	5.715	0.74	8.786	4.986	0.568	16.505	10.701	0.648
rate75_real50	6.313	6.002	0.951	5.555	11.45	2.061	11.868	17.453	1.471
rate50_real50	5.717	6.167	1.079	4.209	9.514	2.261	9.925	15.681	1.58
rate75_real70	6.22	5.798	0.932	8.534	10.397	1.218	14.754	16.195	1.098
rate25_real20	9.487	10.284	1.084	5.907	14.169	2.399	15.394	24.453	1.589
typo30_real0	8.401	7.867	0.936	4.332	4.884	1.128	12.733	12.752	1.001
typo30_real10	9.323	8.488	0.91	6.098	12.244	2.008	15.421	20.732	1.344
typo30_real20	8.179	10.333	1.263	4.745	9.116	1.921	12.924	19.449	1.505
typo30_real30	8.783	6.288	0.716	6.798	8.749	1.287	15.581	15.037	0.965
typo30_real40	9.08	9.385	1.034	6.319	10.195	1.613	15.399	19.58	1.271
typo30_real50	8.859	8.409	0.949	6.803	10.632	1.563	15.663	19.041	1.216
typo30_real60	9.186	6.614	0.72	6.674	7.999	1.199	15.859	14.612	0.921
typo30_real70	9.201	4.47	0.486	6.296	2.257	0.358	15.497	6.727	0.434

LLM judge. The proposal specifies a second, non-lexical measure for both dimensions: a separate model with a fixed, strict prompt. The judge is a different model from the one under test, called at temperature 0 on a seeded random subsample of answered traces per config. It is shown the corrupted-word list (*original* $\rightarrow$ *as shown*), the question, and the reasoning trace (head+tail window). It returns two scalar ratings and evidence: *self_doubt_score* from 0–10, where 0 means no second-guessing and 10 means pervasive uncertainty or looping; and *repair_understanding_score* from 0–5, where 0 means the question’s meaning is fully understood and 5 means the model is fundamentally lost because of the corruption. The judge also returns a one-line justification and verbatim evidence quotes; quotes are checked against the reasoning trace. The judge is a subsample measure, while the lexical tables remain the full-data result.

Repair Behaviour

Table 8: How each corrupted word was handled, per config (categories defined below the table).

config	n_corrupt_words	silent_fix_n	flagged_n	misread_n	not_used_n	silent_fix_frac	flagged_frac	misread_frac	not_used_frac
rate25_real70	3548	2119	216	90	1123	0.5972	0.0609	0.0254	0.3165
rate75_real20	9825	5982	215	142	3486	0.6089	0.0219	0.0145	0.3548
rate50_real20	6964	4463	140	63	2298	0.6409	0.0201	0.009	0.33
rate25_real50	3543	2250	131	78	1084	0.6351	0.037	0.022	0.306
rate75_real40	9776	5621	363	209	3583	0.575	0.0371	0.0214	0.3665
rate50_real60	6932	3995	304	183	2450	0.5763	0.0439	0.0264	0.3534
rate25_real10	3537	2395	50	24	1068	0.6771	0.0141	0.0068	0.302
rate25_real0	3493	2353	21	13	1106	0.6736	0.006	0.0037	0.3166
rate50_real0	6928	4580	46	40	2262	0.6611	0.0066	0.0058	0.3265
rate25_real30	3564	2304	106	49	1105	0.6465	0.0297	0.0137	0.31
rate75_real60	9527	5199	394	308	3626	0.5457	0.0414	0.0323	0.3806
rate50_real40	6989	4239	218	127	2405	0.6065	0.0312	0.0182	0.3441
rate75_real30	9621	5779	253	184	3405	0.6007	0.0263	0.0191	0.3539
rate50_real10	6926	4516	86	42	2282	0.652	0.0124	0.0061	0.3295
rate25_real60	3533	2208	162	83	1080	0.625	0.0459	0.0235	0.3057
rate25_real40	3574	2282	132	55	1105	0.6385	0.0369	0.0154	0.3092
rate75_real0	9490	6100	62	57	3271	0.6428	0.0065	0.006	0.3447
rate75_real10	9839	6215	126	106	3392	0.6317	0.0128	0.0108	0.3448
rate50_real30	6878	4317	180	94	2287	0.6277	0.0262	0.0137	0.3325
rate50_real70	6967	3972	353	199	2443	0.5701	0.0507	0.0286	0.3507
rate75_real50	9793	5563	376	272	3582	0.5681	0.0384	0.0278	0.3658
rate50_real50	7060	4203	263	148	2446	0.5953	0.0373	0.021	0.3465
rate75_real70	9653	5105	466	383	3699	0.5289	0.0483	0.0397	0.3832
rate25_real20	3494	2344	79	28	1043	0.6709	0.0226	0.008	0.2985
typo30_real0	4249	2873	13	13	1350	0.6762	0.0031	0.0031	0.3177
typo30_real10	4247	2846	60	28	1313	0.6701	0.0141	0.0066	0.3092
typo30_real20	4292	2786	83	45	1378	0.6491	0.0193	0.0105	0.3211
typo30_real30	4130	2629	102	55	1344	0.6366	0.0247	0.0133	0.3254
typo30_real40	4244	2651	175	69	1349	0.6246	0.0412	0.0163	0.3179
typo30_real50	4212	2584	168	92	1368	0.6135	0.0399	0.0218	0.3248
typo30_real60	4237	2586	208	96	1347	0.6103	0.0491	0.0227	0.3179
typo30_real70	4183	2516	212	113	1342	0.6015	0.0507	0.027	0.3208

A *corrupted word* is a word that received a typo — it differs between the clean and typo question (e.g. *sum* → *sun*). We recover these by diffing the two questions, then place each into one bucket by which form appears in the reasoning:

- **silent_fix** — only the original word (*sum*) appears: a quiet recovery.
- **flagged** — both forms appear (*sum* and *sun*): noticed and fixed.
- **misread** — only the corrupted form (*sun*) appears: read as the wrong word.
- **not_used** — neither form appears in the reasoning.

Columns ***_n** are counts; ***_frac** are the share of **n_corrupt_words**.

Real-word Effect

This section isolates the effect of typo *kind* (real-word vs non-word), holding the question, corrupted positions, and rate fixed — something the vs-clean accuracy section cannot do. Three views:

- a **controlled paired** comparison of low-real vs high-real variants on the *same* questions, with a McNemar test (paired table);
- a **per-typo logistic regression** comparing the marginal harm of one real-word vs one non-word typo (logit table);
- whether wrong answers **flag the corruption** more or less as the real-word share rises (silent-failure table).

Read the signs, deltas and p -values in the tables below for this dataset’s result.

Lexical Grid (overview)

Table 9: All four marker families (second_guess, uncertainty, typo_noticing, repair_words) across configs.

config	second_guess	uncertainty	typo_noticing	repair_words
clean	7.15	3.47	0.901	0.755
rate25_real70	8.554	7.525	2.486	1.713
rate75_real20	7.57	9.315	3.298	2.434
rate50_real20	8.658	7.904	2.638	1.713
rate25_real50	8.52	7.059	2.084	1.628
rate75_real40	6.252	10.074	3.374	2.316
rate50_real60	7.351	7.634	2.419	1.822
rate25_real10	9.811	6.593	1.959	1.363
rate25_real0	8.854	5.65	1.872	1.459
rate50_real0	7.29	5.949	1.971	1.296
rate25_real30	8.535	5.937	2.007	1.391
rate75_real60	6.626	9.375	2.916	2.132
rate50_real40	7.765	8.121	2.18	1.852
rate75_real30	7.093	6.86	2.675	1.702
rate50_real10	8.421	9.297	2.795	2.156
rate25_real60	8.04	8.137	2.202	1.676
rate25_real40	8.918	8.706	3.14	2.355
rate75_real0	7.124	8.854	2.742	2.064
rate75_real10	8.944	11.834	3.141	2.434
rate50_real30	8.004	7.572	2.365	1.818
rate50_real70	7.044	7.506	2.21	1.798
rate75_real50	6.146	8.73	3.027	2.007
rate50_real50	5.907	6.454	2.337	1.76
rate75_real70	5.984	9.575	2.591	2.009
rate25_real20	9.679	7.895	2.805	2.247
typo30_real0	8.276	4.461	1.502	1.109
typo30_real10	9.146	7.398	1.995	1.43
typo30_real20	8.795	5.995	1.965	1.377
typo30_real30	8.279	7.192	1.989	1.547
typo30_real40	9.161	7.345	2.814	2.071
typo30_real50	8.74	7.814	2.364	1.811
typo30_real60	8.575	6.988	2.022	1.521
typo30_real70	8.189	5.432	2.043	1.402

Spellcheck Recovery